

Motivation

- A unique feature of E-commerce is the possibility of placing inventory away from customers.
- Allows for **risk pooling** via a more **flexible demand fulfillment** and a **joint inventory planning**.
- How to jointly plan inventory across multiple warehouses?
- How to fulfill sequentially arriving customer demand?

Problem Formulation

- An E-commerce business has n **warehouses** and receives demand arriving sequentially at m **customer locations**.
- The business decides:
 - An **initial inventory** $\mathbf{q}$ to hold at each warehouse.
 - An **online fulfillment policy** π of the sequentially arriving customer demand.
- The business incurs:
 - **Inventory costs**: per unit holding cost h for remaining inventory and per unit backlogging cost b for unsatisfied demand at the end of the selling season.
 - **Fulfillment cost**: if x_{ij} units are shipped between i to j , a cost proportional to x_{ij} and the metric distance ℓ_{ij} between i and j is paid.
- The **only information available** to the business is the **mean** and **variance** of the total demand arriving at each location $\mathbf{d}$ and the goal is to **minimize costs in the face of the worst case demand distribution**.
- Formally the business solves the distributionally robust problem,

$$\inf_{\mathbf{q} \geq \mathbf{0}} \inf_{\pi} \sup_{\mathbb{P} \in \mathcal{P}(\boldsymbol{\mu}, \boldsymbol{\Sigma})} \mathbb{E}_{\mathbf{d} \sim \mathbb{P}} C(\mathbf{q}, \pi, \mathbf{d}),$$

where

$$\mathcal{P}(\boldsymbol{\mu}, \boldsymbol{\Sigma}) := \{\mathbb{P} \mid \mathbb{E}_{\mathbf{d} \sim \mathbb{P}}(\mathbf{d}) = \boldsymbol{\mu}, \mathbb{V}_{\mathbf{d} \sim \mathbb{P}}(\mathbf{d}) = \boldsymbol{\Sigma}\}$$

is the set of demand distributions with mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$, and

$$C(\mathbf{q}, \pi, \mathbf{d}) = \sum_{i \in [n], j \in [m]} \ell_{ij} x_{ij} + h \sum_{i \in [n]} \left(q_i - \sum_{j \in [m]} x_{ij} \right) + b \sum_{j \in [m]} \left(d_j - \sum_{i \in [n]} x_{ij} \right)$$

is the cost function.

Related Work

- **Single Location** $n = 1$ with mean μ and std σ : Scarf (1957) [1] gives closed form optimal solution

$$q^{\text{optimal}} = \mu + \frac{\sigma}{2} \left(\sqrt{\frac{b}{h}} - \sqrt{\frac{h}{b}} \right).$$

- **Multi-locations**:
 - Chou et al. (2006) [2] studies the problem under chance-constrained ambiguity set.
 - Govindarajan et al. (2021) [3] studies the problem under moment ambiguity set.
 - Li et al. (2022) [4] studies the problem under scenario-based ambiguity set.
 - etc.

Warm-Up: Uniform Metric $\ell_{ij} = \ell$.

- If the total backlogging cost is ignored, each location i behaves as an independent location with **“virtual” backlogging cost** $\ell - h$ (ℓ for shipping excess inventory from another location and $-h$ gained by the other location from not having to hold the inventory anymore) and holding cost h .
- The **inventory solution** consists of:
 - Holding Scarf’s solution with “virtual” backlog $\ell - h$ and holding cost h at each location.
 - Holding Scarf’s solution with backlog b and holding cost h overall.
- The **fulfillment policy** fulfills from a location with leftover inventory chosen uniformly randomly (similar to the marking algorithm for paging).

General Metric

Well-Separated Hierarchical Partitions.

- Our algorithm begins by computing a special hierarchical clustering of the locations that we denote by $\mathcal{P}$.
- $\mathcal{P}$ is given by a set of clusters beginning from a large cluster that contains all locations and that is recursively refined to smaller clusters until reaching single location clusters.

- $\mathcal{P}$ has the property that it can be partitioned into a **“small”** (polylogarithmic on the number of locations) number of families of clusters $\mathcal{P} = \cup_{k=1}^K \mathcal{P}_k$ where each family $\mathcal{P}_k$ consists of a set of **“well-separated” clusters** (separated by some ℓ_k large enough compared to the diameter of the clusters).

Inventory Solution.

- Each family $\mathcal{P}_k$ can be thought of as an instance with uniform metric with $\ell_{ij} = \ell_k$.
- The inventory solution consist of:
 - Holding Scarf’s solution with “virtual” backlog $\ell_k - h$ and holding cost h at each cluster of $\mathcal{P}_k$.
 - Holding Scarf’s solution with backlog b and holding cost h overall.

Fulfillment Policy.

- Demand arrives at a location i .
- Let C be the “smallest” cluster containing i that still has inventory left.
- To fulfill the demand, choose a subcluster of C with remaining inventory uniformly randomly. Within the chosen subcluster choose again one of its subclusters uniformly randomly and so on until a location is chosen. Fulfill demand from the chosen location.

Main Theorem. Our policy gives a polylogarithmic (in the number of locations) approximation to the problem.

Our policy also shows a **strong empirical performance** on extensive numerical experiments we conducted.

References

- [1] Herbert E Scarf. A Min-Max Solution of an Inventory Problem. RAND Corporation, 1957.
- [2] Mabel Chou, Melvyn Sim, and Rick So. “A robust optimization framework for analyzing distribution systems with transshipment”. In: Technical Report; Working paper. School of Business, National University of Singapore Singapore, 2006.
- [3] Aravind Govindarajan, Amitabh Sinha, and Joline Uichanco. “Distribution-free inventory risk pooling in a multilocation newsvendor”. In: Management Science 67.4 (2021), pp. 2272–2291.
- [4] Chenxi Li et al. “Distributionally robust multilocation newsvendor at scale: A scenario-based linear programming approach”. In: Available at SSRN 4207042 (2022).